



N¹-methylpseudouridine-incorporated mRNA outperforms pseudouridine-incorporated mRNA by providing enhanced protein expression and reduced immunogenicity in mammalian cell lines and mice



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ARTICLE INFO

Article history:

Received 1 June 2015

Received in revised form 17 August 2015

Accepted 26 August 2015

Available online 3 September 2015

Chemical compounds studied in this article:

N¹-methylpseudouridine (PubChem CID: 99543)

pseudouridine (PubChem CID: 15047)

5-methylcytidine (PubChem CID: 92918)

Keywords:

mRNA

Modified RNA

Nucleobase modifications

Antiviral innate immunity

Toll-like receptor

Gene therapy

ABSTRACT

Messenger RNA as a therapeutic modality is becoming increasingly popular in the field of gene therapy. The realization that nucleobase modifications can greatly enhance the properties of mRNA by reducing the immunogenicity and increasing the stability of the RNA molecule (the Kariko paradigm) has been pivotal for this revolution. Here we find that mRNAs containing the N¹ methylpseudouridine (m1Ψ) modification alone and/or in combination with 5 methylcytidine (m5C) outperformed the current state of the art pseudouridine (Ψ) and/or m5C/Ψ modified mRNA platform by providing up to ~44 fold (when comparing double modified mRNAs) or ~13 fold (when comparing single modified mRNAs) higher reporter gene expression upon transfection into cell lines or mice, respectively. We show that (m5C/m1Ψ) modified mRNA resulted in reduced intracellular innate immunogenicity and improved cellular viability compared to (m5C/Ψ) modified mRNA upon in vitro transfection. The enhanced capability of (m5C/m1Ψ) modified mRNA to express proteins may at least partially be due to the increased ability of the mRNA to evade activation of endosomal Toll like receptor 3 (TLR3) and downstream innate immune signaling. We believe that the (m5C/m1Ψ) mRNA platform presented here may serve as a new standard in the field of modified mRNA based therapeutics.

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1. Introduction

Messenger RNA as a gene expression platform has numerous advantages over pDNA based expression modalities. Unlike pDNA, mRNA does not need to enter the nucleus to carry out its function. Thus, mRNAs can immediately express proteins inside a cell (even in cells that are not rapidly dividing) [1]. mRNA vectors are also safer than pDNA vectors as expression from mRNA is transient. Furthermore, mRNA poses virtually no risk of genomic integration and mutagenesis of critical regions of the host genome.

While the concept of using mRNA for protein replacement therapy had been originally demonstrated 25 years ago [2], this approach initially did not gain popularity due to the general instability and immunogenicity of the RNA molecule. However, a study conducted by Kariko's

group provided a breakthrough in the field of mRNA therapy by demonstrating that the incorporation of base modifications found in natural RNAs such as m5C, N⁶ methyladenosine (m6A), Ψ, 5 methyluridine (m5U), and 2 thiouridine (s2U) either separately or in combination into mRNA can reduce the TLR mediated immunogenicity of mRNA [3]. Subsequently, it was shown that the Ψ modification in particular could increase the translational capacity and biological stability of mRNA [4]. The increased translational capacity of Ψ modified RNA was later shown to be aided by (1) diminished activation of protein kinase R (PKR) by the modified mRNA and reduced PKR mediated phosphorylation of the α subunit of eukaryotic translation initiation factor 2 (eIF2α) [5,6], (2) reduced activation of 2' 5' oligoadenylate synthetase (OAS) by the modified mRNA and reduced 2' 5' oligoadenylate mediated activation of RNase L [7] and (3) increased resistance of the modified mRNA to RNase L mediated degradation [7].

Following the development of the modified mRNA platform, various pre clinical studies have demonstrated its vast potential for therapeutic applications including the work of Warren et al. [8] which used m5C/Ψ

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modified mRNA to reprogram and differentiate human cells, the work of Miki, Endo et al. [9] which used m5C/ Ψ modified mRNA as a selection switch to enrich for specific types of cells following stem cell differentiation, the work of Wroblewska et al. [10] which used m5C/ Ψ modified mRNA to create a synthetic gene circuit that selectively kills cancer cells, the work of Kormann et al. [11] which used m5C/s2U modified mRNA to treat mice with surfactant protein B (SP B) deficiency (a lethal congenital lung disease), the work of Zangi, Lui et al. [12] which used m5C/ Ψ modified mRNA to treat a mouse model of myocardial infarction, and the work of Mahiny et al. [13] which used m5C/s2U modified mRNA to express zinc finger nucleases in mice for genome editing.

Given the pre clinical success in using mRNA with various modified bases for therapeutic applications in vitro and in vivo, we sought to identify RNA base modifications that could further reduce the immunogenicity and increase the translational capacity of mRNA.

Here we demonstrate that m1 Ψ , a modification naturally found in 18S rRNA as a precursor of N¹ methyl N³ (3 amino 3 carboxypropyl)pseudouridine (m1acp3 Ψ) [14], drastically improved the translational capacity of mRNA when incorporated alone or in combination with m5C compared to (m5C)/ Ψ modified RNA in A549 cells (human lung epithelial cells), BJ cells (human foreskin fibroblasts), C2C12 cells (murine myoblasts), HeLa cells (human cervix epithelial cells), human primary keratinocytes from neonatal foreskin, as well as in mice after intradermal (i.d.) or intramuscular (i.m.) injection of the modified RNA. We show in the various cell lines that (m5C)/m1 Ψ modified mRNA had reduced cytotoxicity compared to (m5C)/ Ψ modified mRNA. (m5C)/m1 Ψ modified mRNA also showed reduced activation of the intracellular innate immune response. Finally, we show that the superiority of (m5C)/m1 Ψ modified mRNA over (m5C)/ Ψ modified mRNA may at least partially be due to an increase in its ability to avoid TLR3 activation. Thus, we believe that (m5C)/m1 Ψ modified mRNA may replace (m5C)/ Ψ modified mRNA as the new standard for mRNA based therapeutic applications.

2. Materials and methods

2.1. Cells and reagents

Human lung epithelial cells (A549, ATCC® CL 185™), human foreskin fibroblasts (BJ, ATCC® CRL 2522™), murine muscle cells (C2C12, ATCC® CRL 1772™), human cervix epithelial cells (Hela, ATCC® CCL 2™), and human primary keratinocytes from neonatal foreskin (ATCC® PCS 200 010™) were purchased from ATCC and cultured according to the conditions recommended by ATCC. Human embryonic kidney cells (HEK293) and HEK293 cells stably overexpressing TLR3 (HEK293 TLR3 cells) were a generous gift from Prof. Rudi Beyaert (Vlaams Instituut voor Biotechnologie). Modified nucleoside triphosphates (5 methylcytidine triphosphate, pseudouridine triphosphate, and N¹ methylpseudouridine triphosphate) were purchased from TriLink Biotechnologies. Lipofectamine 2000 (Lipo 2 K) was purchased from Invitrogen. Potassium 2 (6 oxo 1,3 benzothiazol 2 ylidene) 4 thiazolidinecarboxylate (D luciferin, potassium salt), for in vivo measurement of firefly luciferase activity, was purchased from Gold Biotechnology.

2.2. Plasmid

The plasmid (pTK305) used for in vitro transcription of firefly luciferase encoding mRNA was constructed using standard cloning procedures including In Fusion PCR cloning (Clontech) and Gateway cloning (Invitrogen). The plasmid included a bacteriophage T7 polymerase promoter, the open reading frame (ORF) of interest (firefly luciferase) flanked by the 5' UTR of the *Venezuelan equine encephalitis virus* (VEEV) subgenomic RNA and two tandem repeats of the 3' UTR of VEEV subgenomic RNA, a 40 nucleotide poly(A) sequence, and

consensus recognition sequences for the I SclI homing endonuclease upstream of the T7 promoter and downstream of the poly(A) sequence. The *E. coli* glycerol stock, sequence, and map for the plasmid used in this study have been deposited to Addgene (www.addgene.org/Ron_Weiss/).

2.3. mRNA in vitro production

mRNA was produced by in vitro transcription (IVT) of I SclI linearized plasmid DNA using the MEGAScript® T7 Transcription Kit (Invitrogen) with unmodified nucleotides or a combination of the modified nucleotides described above. For IVT of modified mRNA, the modified nucleotides entirely replaced unmodified nucleotides in the reaction. Following IVT, the RNA was treated with DNase I (Invitrogen) and purified using the RNeasy® Mini Kit (Qiagen), denatured at 65 °C, enzymatically (cap1) capped using the ScriptCap™ m7G Capping System (Cellscript) and ScriptCap™ 2' O Methyltransferase Kit (Cellscript), poly(A) tailed using the A Plus™ Poly(A) Polymerase Tailoring Kit (Cellscript), and re purified using the RNeasy® Mini Kit (Qiagen) following the manufacturers' protocols.

2.4. In vitro mRNA lipofection and firefly luciferase assay

mRNA was mixed with Lipo 2 K at a ratio of 1:2 (µg mRNA: µl Lipo 2 K) in Opti MEM I (Invitrogen). The complexes were allowed to form for 30 min at room temperature. The average hydrodynamic size and zeta potential of the RNA/lipid complexes were determined as described previously [1] by dynamic light scattering and laser Doppler electrophoresis using a Zetasizer Nano ZS (Malvern). Transfections were performed by adding 1 µg of complexed mRNA to cells pre seeded in 24 well plates. The complexes were removed 4 h later along with Opti MEM I and were replaced with the ATCC recommended culture media containing serum. 24 h after transfection, cells were lysed with 100 µl of 1 × Passive Lysis Buffer (Promega) and firefly luciferase activity was measured with the Luciferase Assay Kit (Promega) according to the manufacturer's protocol. Bioluminescence was measured using the GloMax luminometer (Promega).

2.5. Protein concentration assay

Protein concentrations of cell lysates were measured using the Pierce™ Micro BCA™ Protein Assay kit (Thermo Scientific). The standard manufacturer's protocol was followed after lysing the cells with Passive Lysis Buffer and diluting the lysate (1:10) in water.

2.6. Viability assay

Mammalian cells were transfected in 24 well plates with 1 µg of unmodified or modified RNA as described above. 24 h after transfection, the viability of mRNA transfected cells was measured using an MTT proliferation assay according to the manufacturer's protocol (Roche).

2.7. Enzyme linked immunosorbent assay (ELISA)

Cell culture supernatants were collected 24 h after transfection with mRNA and stored at –80 °C until the ELISAs were performed unless stated otherwise. The ELISA kits for human interferon β (IFN β) and chemokine (C C motif) ligand 5 (CCL5; also known as RANTES) were purchased from BioLegend and Life Technologies, respectively. ELISAs were performed according to the manufacturers' recommendations, as described previously [15].

2.8. Immunostaining of TLR3 and flow cytometry

24 h after transfection with mRNA, cells were collected, washed with phosphate buffered saline without calcium or magnesium (DPBS, no calcium, no magnesium; Gibco) and incubated at room temperature

for 1 h in 1 × Fixation Buffer (eBioscience). The fixed cells were washed twice with 1 × Permeabilization Buffer (eBioscience), resuspended in 1 × Permeabilization Buffer and incubated in the dark for 30 min with Phycoerythrin (PE) conjugated anti TLR3 antibodies (BioLegend). The cells were then washed twice with PBS to get rid of any unbound antibodies, and resuspended in PBS. Fluorescence signal was measured by flow cytometry on a BD Accuri™ C6 flow cytometer (BD Biosciences). Data were analyzed using the CFlow Plus Analysis software (BD Biosciences). Live cells were gated based on the areas of the forward/side scatters and single cells were gated based on the width of the forward scatter. The mean fluorescence intensity (MFI) of the PE dye (excitation laser: 488 nm, emission filter: 585/40 nm) was calculated based on the live, single cell population.

2.9. Mouse experiments

7 week old Balb/c mice were obtained from Janvier. Mice were housed in individually ventilated cages under 12:12 h dark/light cycle conditions. Access to food and water was maintained ad libitum. All experiments were carried out with the approval of the Ghent University Ethics Committee (No. EC 2014/57). Mice were anesthetized with constant flow of isoflurane during intradermal (i.d.) or intramuscular (i.m.) injections. 20 µg of mRNA were complexed with Lipo 2 K at a 1:1 (µg mRNA: µl Lipo 2 K) ratio and resuspended in PBS and injected i.d. or i.m. into the tibialis anterior muscle.

2.10. In vivo imaging of firefly luciferase expression

The expression levels of firefly luciferase in mice were measured over time using the in vivo bioluminescent imaging system, IVIS Lumina II (PerkinElmer), until the signal from the injected mRNA reached background levels. Mice were injected intraperitoneally (i.p.) with 50 mg/kg of D luciferin and bioluminescence was measured 10 min after the injection. Acquisition settings were set at f stop: 1, binning: 8, and auto exposure.

2.11. Statistical analysis

The data points are represented as mean ± standard deviation (SD) unless stated otherwise. Statistical analysis was performed using

GraphPad Prism 6 (GraphPad Software). One way ANOVA and Tukey's multiple comparisons tests were used to identify statistically significant differences among different mRNA groups. Differences were considered significant when $p \leq 0.05$.

3. Results

3.1. (m5C)/m1Ψ modified mRNA has a higher translational capacity than (m5C)/Ψ modified mRNA when lipofected in vitro

In order to test whether there are nucleobase modifications found in natural RNAs that are superior to Ψ at enhancing the translational capacity of mRNA, we incorporated m1Ψ into RNA by in vitro transcription. Ψ and m1Ψ are natural derivatives of uracil that can be distinguished by the N¹ positions of their bases (m1Ψ is methylated at N¹) (Fig. 1A–C).

In addition to Ψ or m1Ψ single modified mRNA, we decided to compare m5C/Ψ or m5C/m1Ψ double modified mRNAs, since it had been shown previously that m5C (a natural derivative of cytosine; see Fig. 1D,E) can increase the translational capacity of Ψ single modified mRNA [8]. We lipofected unmodified or modified mRNAs encoding the firefly luciferase gene into several cell lines: A549 cells (human lung carcinoma cells), BJ cells (human foreskin fibroblasts), C2C12 cells (mouse myoblasts), and HeLa cells (human cervical adenocarcinoma cells) as well as primary cells (human neonatal foreskin primary keratinocytes). We chose cell lines or primary cells of different cell types or those derived from diverse organisms to ensure that we would be able to generalize the effects that we observed. 24 h after mRNA transfection into the various cell lines and primary cells, luciferase assays were performed. As shown in Fig. 2 and Supplementary Table 1, we observed statistically significant differences in luciferase activity in all of the cell types transfected with the differentially modified mRNAs. Specifically, the m5C/m1Ψ double modified mRNA resulted in the highest luciferase activity in all cell types except C2C12 myoblast cells. In these myoblast cells, the single m1Ψ modified mRNA gave the highest luciferase activity. The observation that the translational effects that the various modifications had on the different types of cells were distinct is in conceptual agreement with a previous study by Uchida et al. which compared the effects of m5C, Ψ, m5C/s2U, m5C/Ψ, and m5C/s2U/Ψ modified mRNA on protein expression in various cell

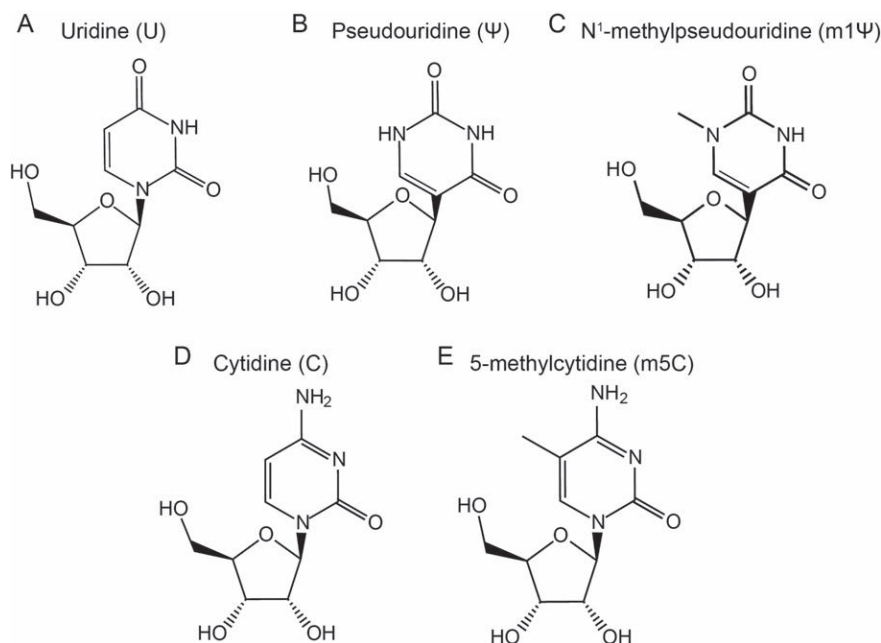


Fig. 1. Chemical structures of modified nucleosides used in this study. The chemical structures of (A) uridine, (B) Ψ, (C) m1Ψ, (D) cytidine, and (E) m5C. Chemical structures were adapted from the MODOMICS database [16].

Luciferase assay, lipofection, 24 h

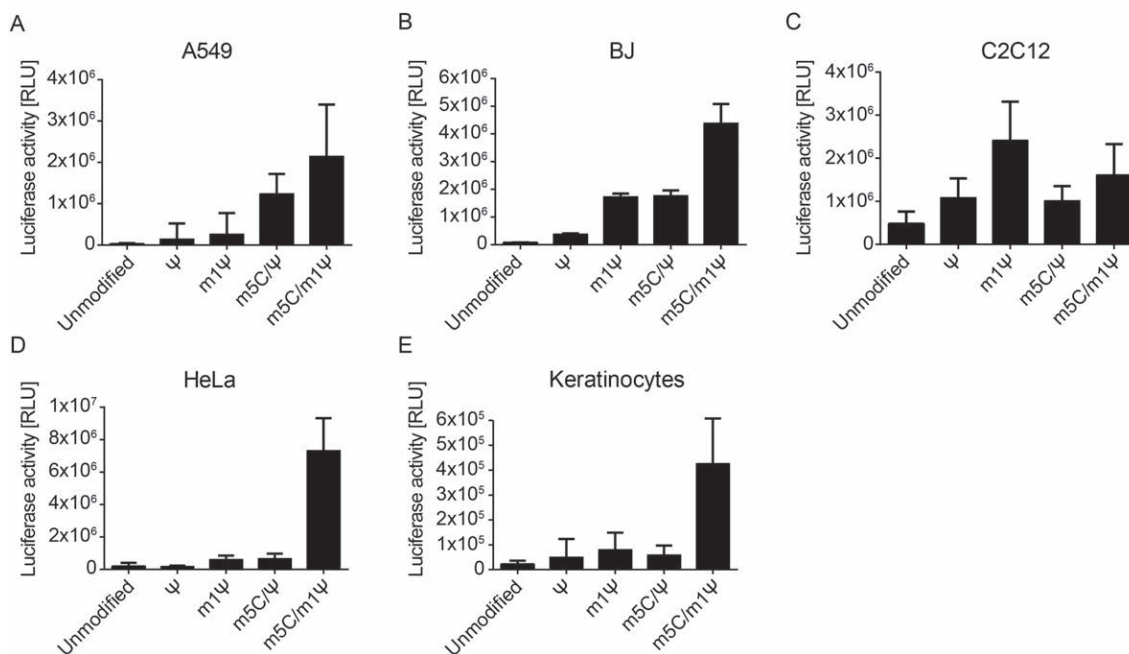


Fig. 2. Comparison of luciferase activity 24 h after lipofection of unmodified or modified (Ψ , m1 Ψ , m5C/ Ψ , m5C/m1 Ψ) mRNAs encoding firefly luciferase in various cell lines. Luciferase activities for each mRNA species are shown for (A) A549 cells, (B) BJ cells, (C) C2C12 cells, (D) HeLa cells, and (E) human neonatal foreskin primary keratinocytes. 1 μ g of each mRNA species was transfected into each cell type by lipofection. The results are presented as the mean $\pm$ SD ($n = 3$, one-way ANOVA and Tukey's multiple comparisons test results in Supplementary Table 1).

types [17]. The differences in the expression levels that we observed between the unmodified and modified mRNAs were not due to physical differences in the various RNA/lipid complexes as the mean particle size and zeta potential of the particles containing the unmodified and modified mRNAs were similar (Supplementary Table 2). Therefore, we demonstrate that m1 Ψ or m5C/m1 Ψ modified mRNA outperformed the previous state of the art Ψ or m5C/ Ψ modified mRNA expression platform, respectively.

3.2. The translational lifetime of m5C/m1 Ψ modified mRNA is longer than that of m5C/ Ψ modified mRNA when lipofected in vitro

We next assessed the duration of protein expression after lipofection of A549 cells with unmodified or modified mRNAs. We measured the luciferase activity 3, 6, 12, 24, and 48 h after transfection. As shown in Fig. 3, there was a statistically significant difference in luciferase production from the various mRNAs ($p = 0.017$ by two way repeated measures ANOVA). At each time point, the m5C/m1 Ψ modified mRNA outperformed the rest of the mRNAs. We observed an initial burst of luciferase activity at the 3 h time point, where m5C/m1 Ψ modified mRNA produced ~916.7 fold more activity than unmodified mRNA, ~118 fold more than Ψ modified mRNA, 23 fold more than m1 Ψ modified mRNA and 44 fold more than m5C/ Ψ modified mRNA. Subsequently, the expression from the mRNAs dropped drastically between the 3 and 6 h time points after which the drop in luciferase activity was less severe. Importantly, the relative luciferase signal between the 24 and 48 h time points decreased the least for the m5C/m1 Ψ modified mRNA. Thus m5C/m1 Ψ outperformed the m5C/ Ψ modified mRNA with regards to duration of expression.

3.3. (m5C)/m1 Ψ modified mRNA is less cytotoxic than (m5C)/ Ψ modified mRNA when lipofected in vitro

We previously demonstrated that the transfection of mRNAs produced by IVT into mammalian cells can decrease the viability of

the transfected cells [15]. Since m5C/ Ψ modified mRNA caused drastically less cell death upon transfection relative to unmodified mRNA [15], we next sought to determine how the cytotoxicity of m1 Ψ modified mRNA compared with that of Ψ modified mRNA. For this, we lipofected the mRNAs into the cell lines mentioned above and performed MTT assays to quantify the amount of viable cells. As shown in Fig. 4, the effects of the various mRNAs on cell viability were dependent on the cell type, where all cell types except primary keratinocytes showed a statistically significant difference in the overall pattern of

Luciferase assay, lipofection, A549 cells

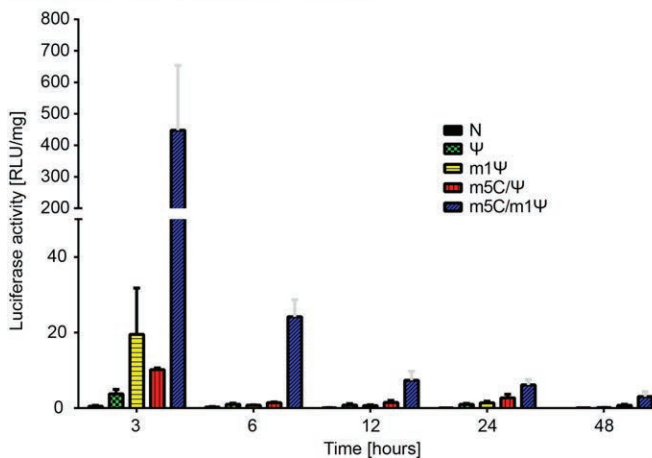


Fig. 3. Kinetics of luciferase activity after lipofection of unmodified or modified (Ψ , m1 Ψ , m5C/ Ψ , m5C/m1 Ψ) mRNAs encoding firefly luciferase in A549 cells. Luciferase activities were measured for each mRNA species at 3, 6, 12, 24, and 48 h time points. Luciferase activity was normalized to the amount of total cellular protein measured by a BCA assay to correct for the differences in cell viability at the different time points. 1 μ g of each mRNA species was transfected. The results are presented as the mean $\pm$ SD ($n = 3$).

MTT assay, lipofection, 24 h

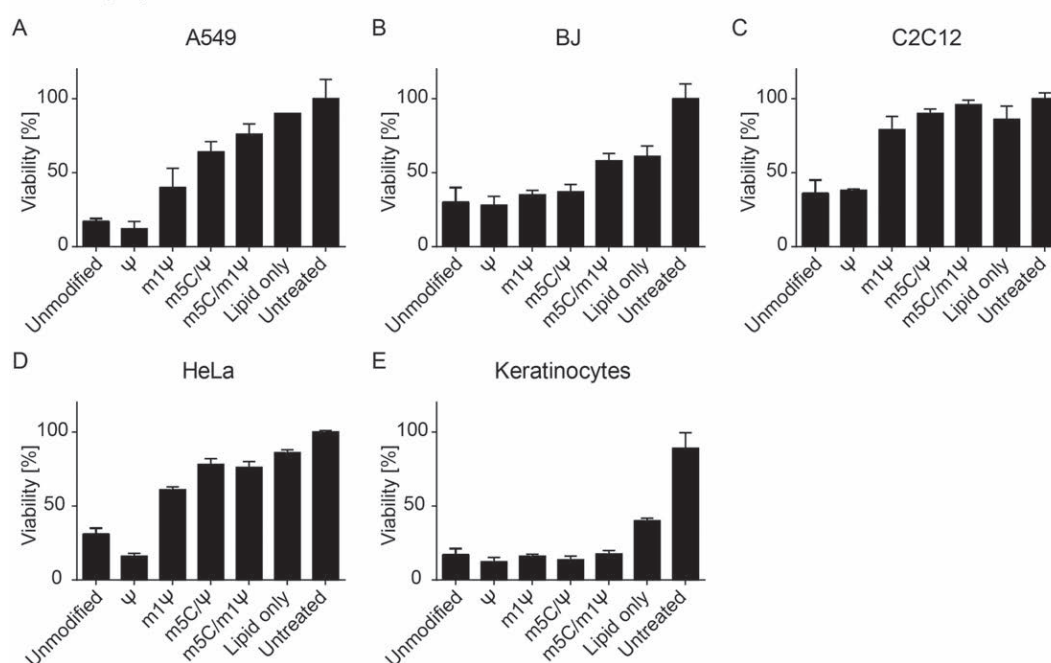


Fig. 4. Viability of mammalian cells 24 h after lipofection of unmodified or modified (Ψ , m1 Ψ , m5C/ Ψ , m5C/m1 Ψ) mRNAs determined using an MTT assay. 1 μ g of each mRNA species was transfected into (A) A549, (B) BJ, (C) C2C12, (D) HeLa, and (E) primary keratinocytes by lipofection. The results are presented as the mean $\pm$ SD ($n = 3$).

viability ($p \leq 0.0001$ for A549, $p \leq 0.01$ for BJ, $p \leq 0.0001$ for C2C12, $p \leq 0.0001$ for HeLa, and $p > 0.05$ for keratinocytes by one way ANOVA). Specifically, unmodified and Ψ modified mRNA were significantly more toxic than the other modified mRNAs in all cells except keratinocytes. In C2C12 and HeLa cells, m1 Ψ modified or double modified (m5C/ Ψ or m5C/m1 Ψ) mRNAs did not cause additional toxicity when compared to cells treated with Lipo 2 K. In BJ cells, only the m5C/m1 Ψ modified mRNA was non toxic. The results obtained using keratinocytes may have been elicited by the toxicity of Lipo 2 K or the lipofection process in these cells. Nevertheless, in summary we find that while the cytotoxicity of IVT mRNA was dependent on the cell type, when we observed a difference in cellular viability upon transfection of the various mRNAs, m1 Ψ or m5C/m1 Ψ was less cytotoxic than Ψ or m5C/ Ψ modified mRNA, respectively.

3.4. (m5C)/m1 Ψ modified mRNA stimulates intracellular innate immune signaling pathways less than (m5C)/ Ψ modified mRNA when lipofected in vitro

Since the superior translational capacity and reduced cytotoxicity of modified mRNAs are generally known to correlate with reduced activation of intracellular innate immune pathways [3,15], we next asked whether there was a difference in the activation of key cytokines upon transfection of the differentially modified mRNAs. For this, we lipofected A549 cells with the various mRNAs and measured by ELISA the levels of secreted IFN β (the master orchestrator of intracellular innate immune signaling) and CCL5/RANTES (a key chemokine previously shown to be most significantly upregulated among TLR pathway genes in A549 cells upon mRNA transfection [15]). As shown in Fig. 5, we find that there was a statistically significant difference in the secreted levels of IFN β and CCL5/RANTES (both $p \leq 0.0001$ by one way ANOVA). Specifically, the IFN β production from cells transfected with Ψ , m1 Ψ , m5C/ Ψ and m5C/m1 Ψ modified mRNAs were respectively reduced by ~ 3 , 11, 4, and 14 fold relative to unmodified mRNA (Fig. 5A). Strikingly, lipofection with m1 Ψ containing mRNAs did not result in significant induction of IFN β . For CCL5/RANTES, as shown in Fig. 5B, the m5C/ Ψ

and m5C/m1 Ψ double modified mRNAs did not induce secretion of CCL5/RANTES. For the single modified mRNAs, m1 Ψ modified mRNA induced much less secretion of CCL5/RANTES than Ψ modified mRNA. Thus, overall, m1 Ψ and/or m5C/m1 Ψ were respectively less immunogenic than Ψ and/or m5C/ Ψ when assessed by the extent of IFN β or CCL5/RANTES secretion.

3.5. TLR3 overexpression converts HEK293 cells into preferential expressers of m1 Ψ modified mRNA over Ψ modified mRNA

We next sought to address the mechanism by which cells preferentially translate (m5C)/m1 Ψ modified mRNA over (m5C)/ Ψ modified mRNA. Since base modifications are known to reduce the activation of the intracellular innate immune response by evading TLR signaling [3, 4], we asked whether differential signaling through TLRs could explain the difference in the translation levels of (m5C)/m1 Ψ and (m5C)/ Ψ modified mRNA. To test this hypothesis, we took advantage of the HEK293 cell line, which normally does not express endosomal TLRs (Supplementary Fig. 1). We lipofected unmodified and modified mRNAs encoding luciferase into normal HEK293 and HEK293 TLR3 cells (which stably overexpress TLR3). In normal HEK293 cells, we did not observe a statistically significant difference in luciferase activity between the modified RNAs (Fig. 6 and Supplementary Table 3). However, strikingly, in HEK293 TLR3, we observed a difference in the pattern of luciferase activity where m1 Ψ modified mRNA exhibited ~ 6 fold more luciferase activity compared to Ψ modified mRNA (Fig. 6 and Supplementary Table 3). In contrast, we did not observe a difference in luciferase expression between m5C/m1 Ψ and m5C/ Ψ modified mRNAs when they were transfected into HEK TLR3 cells. These contextual effects of modifications on protein expression that we found are conceptually consistent with the observations that we had made in Fig. 2 (that the specific effects that modifications have on protein expression are dependent on the cell type) as well as with a previous study by Uchida et al. [17]. Nevertheless, the data are consistent with the hypothesis that reduced activation of TLR3 signaling enables m1 Ψ modified mRNA to translate more protein compared with Ψ modified mRNA.

Cytokine ELISA, lipofection, A549 cells, 24 h

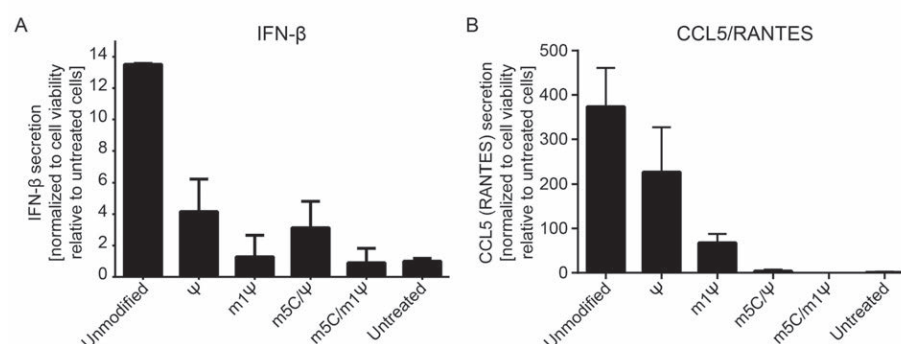


Fig. 5. Secreted cytokine levels (relative to untreated cells) 24 h after lipofection of unmodified or modified (Ψ , m1 Ψ , m5C/ Ψ , m5C/m1 Ψ) mRNAs into A549 cells. 1 μ g of each mRNA type was lipofected into A549 cells and the supernatants were subjected to ELISA to detect (A) IFN- β or (B) CCL5/RANTES. The results are presented as the mean $\pm$ SD ($n = 3$).

3.6. (m5C/) $m1\Psi$ modified mRNA has a higher translational capacity than (m5C/) Ψ modified mRNA when lipofected in mice

Finally, we assessed whether (m5C/) $m1\Psi$ modified mRNA could also produce more protein compared with (m5C/) Ψ modified mRNA when injected in vivo, into mice. For this, we lipofected mRNAs encoding firefly luciferase by i.d. or i.m. injection into mice. The kinetics of expression were followed by bioluminescent imaging (BLI) over a period of 21 days. As shown in Fig. 7A,B, as expected, luciferase activity decayed over time. Importantly, as shown in Fig. 7C,D, the RNA modifications affected the total amount of luciferase that was expressed in vivo (as measured by quantifying the area under the curve of each series in Fig. 7A,B). Unexpectedly, single modified mRNAs (Ψ and m1 Ψ) translated drastically better than their double modified mRNA counterparts (m5C/ Ψ and m5C/m1 Ψ , respectively). Furthermore, m5C/ Ψ modified mRNA exhibited less luciferase activity compared with unmodified mRNA as well as m5C/m1 Ψ modified mRNA. Most notably, m1 Ψ modified mRNA outperformed all other combinations of modifications and translated ~13 fold more than Ψ modified mRNA. Thus, in conclusion, m1 Ψ and m5C/m1 Ψ modified mRNA respectively have a higher translational capacity than Ψ and m5C/ Ψ modified mRNA in vivo.

4. Discussion

A decade has passed since Kariko et al. discovered that incorporation of modified bases into mRNA can reduce the intracellular innate immunogenicity of in vitro transcribed RNA [3]. The subsequent demonstration that base modifications enhance the translational capacity and

stability of RNA [4] inspired a series of successful pre clinical studies in which mRNAs with different combinations of modified bases were used for various therapeutic applications [8,13,18,20].

In this manuscript we sought to identify naturally existing base modifications that may further improve the translational capacity and reduce the immunogenicity of mRNA beyond the current state of the art (m5C/) Ψ modified mRNA platform. Indeed, we found that (m5C/) $m1\Psi$ modified mRNA increased reporter gene expression by more than an order of magnitude relative to (m5C/) Ψ modified mRNA in multiple cell lines and in mice. This may at least partially be explained by the ability of the mRNA to more effectively evade endosomal TLRs such as TLR3.

However, the drastic enhancement of luciferase activity for the single m1 Ψ modified mRNA compared with the arguably stealthier double m5C/m1 Ψ modified mRNA upon lipofection in vivo cannot be explained by the capacity of the RNAs to evade pattern recognition receptors (PRRs). One explanation for this could be that the RNAs lipofected in vivo did not ultimately end up in a cellular context that would cause the RNAs to intensely stimulate the intracellular innate immune response. For instance, the majority of the RNA molecules lipofected in vivo may have been trapped in inactive subcellular compartments such as P bodies inside the cell. Alternatively, PRRs may not be highly expressed in the myofibers or cells in the skin that took up the RNA in mice and thus the incorporation of m1 Ψ alone may have enabled the RNA to be stealthy enough to completely evade any PRRs that reside in these tissues. In such situations, where either very few mRNA molecules are being translated and/or exposed to PRRs, the observed differences in luciferase activity may be determined predominantly by the rate of translation initiation/elongation of the mRNAs (instead of their reduced ability to stimulate the intracellular innate immune

Luciferase assay, lipofection, 24 h

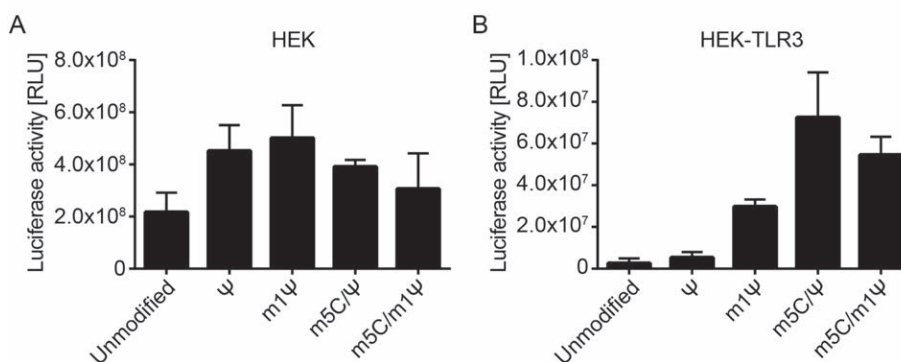


Fig. 6. Comparison of luciferase activity 24 h after lipofection of unmodified or modified (Ψ , m1 Ψ , m5C/ Ψ , m5C/m1 Ψ) mRNAs encoding firefly luciferase in (A) HEK293 and (B) TLR3 overexpressing HEK293 (HEK293-TLR3) cells. 1 μ g of each mRNA species was transfected by lipofection. The results are presented as the mean $\pm$ SD ($n = 3$, one-way ANOVA and Tukey's multiple comparisons test results in Supplementary Table 3).

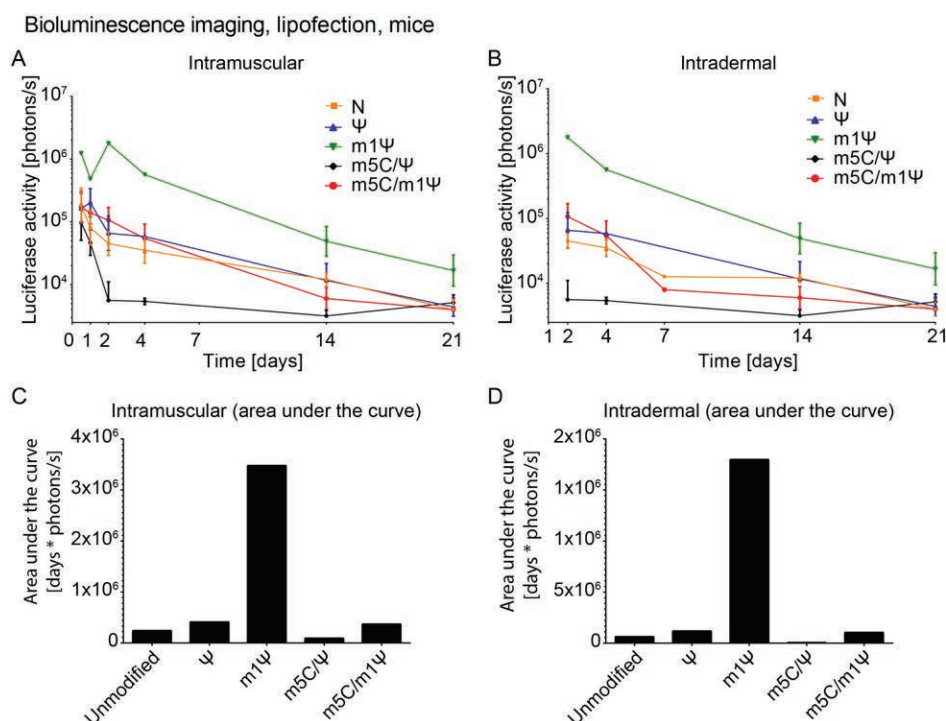


Fig. 7. Comparison of in vivo luciferase activity following injection of unmodified (N) or modified (Ψ , m1 Ψ , m5C/ Ψ , m5C/m1 Ψ) mRNAs administered through i.m. or i.d. routes by lipofection. Luciferase activities measured by BLI for each mRNA species over the course of 21 days for (A) i.m. and (B) i.d. delivery routes. The results are presented as the mean $\pm$ SD ($n \geq 4$). The area under the curve of each series was calculated and plotted to estimate the total amount of protein expression per series for (C) i.m. and (D) i.d. delivery. 20 μ g of each mRNA species was administered to mice for each delivery method and route.

response). Either of these scenarios (in which m1 Ψ alone would completely suppress innate immunity) could explain the unexpected results that we observed in vivo as it had indeed been recently shown that double modified mRNA has a slower translation initiation and/or elongation rate compared to unmodified or single modified mRNA in rabbit reticulocyte lysates [17].

We showed that (m5C)/m1 Ψ modified mRNA had reduced cytotoxicity and immunogenicity compared to (m5C)/ Ψ modified mRNA. The reduction in intracellular innate immune signaling would be advantageous for using mRNA for protein replacement therapy, however, in contrast, it may not be optimal for the purpose of mRNA vaccination, where inflammatory cytokines are known to enhance the adaptive immune response. Therefore, for mRNA vaccination, it may be necessary to identify an optimal balance between mRNA translation and innate immune stimulation by fine tuning the levels of base modifications. Alternatively, instead of inducing innate immune stimulating cytokines by activating PRRs with RNA, cytokines could be heterologously expressed during vaccination from the modified mRNAs themselves.

The use of modified nucleosides and nucleotides that are not naturally incorporated into the RNAs of natural organisms have been explored for the purpose of antiviral therapy [21]. Such “unnatural” chemical base modifications could also be, in theory, used to enhance the properties of mRNA. However, great safety precautions must be taken when doing so as the administration of unnatural modified nucleosides into human patients had previously resulted in mitochondrial toxicity, liver failure, and even death of patients during clinical trials [22]. Furthermore, unlike their native counterparts, unnatural modifications may cause the therapeutic mRNA to induce an adaptive immune response.

Therefore, a more cautious strategy may be to only use base modifications that are naturally found in RNAs for therapeutic applications. Currently, 66 nucleoside modifications have been demonstrated to post transcriptionally exist in eukaryotic RNAs (51 are found in tRNAs, 23 in rRNAs, 13 in mRNAs, and 11 in snRNAs [23]). The current state of the art mRNA enhancing modification Ψ is the most prevalent

nucleoside modification found in nature and was originally thought to be only incorporated into tRNAs, rRNAs, and snRNAs, mainly to stabilize the structures of the RNAs [24]. However, recent genome wide mapping studies have demonstrated that Ψ is also naturally incorporated into mRNAs as well as snoRNAs [25,26]. Incorporation of Ψ into mRNA was upregulated by cellular stress conditions such as heat shock or nutrient deprivation, thus implicating Ψ as a possible native regulator of mRNA function.

More recently, it had been shown that m6A, the most highly abundant base modification found in mRNAs, can indeed play such a regulatory role by affecting the secondary structure of mRNAs [27,28] and/or by directly promoting [29–31] or indirectly inhibiting [28] the binding of RNA binding proteins to decrease the stability of mRNAs [28,29,32], increase the rate of translation initiation [31], or affect processes such as RNA splicing [28], microRNA processing [30], and stem cell differentiation [32].

In contrast, the function of m1 Ψ , a precursor of m1acp3 Ψ in 18S rRNA [14], has not been well characterized. Whether m1 Ψ is naturally incorporated into native cellular mRNAs and whether it could play a regulatory role remains to be determined. Nevertheless, future studies may address the mechanisms by which (m5C)/m1 Ψ modified mRNA provides further enhanced translation and reduced immunogenicity compared to (m5C)/ Ψ modified mRNA. Our results indicated that (m5C)/m1 Ψ modified mRNA may be able to evade endosomal TLRs more efficiently than (m5C)/ Ψ . However, it is also possible that (m5C)/m1 Ψ modified mRNA can evade retinoic acid inducible gene I (RIG-I) like receptors or PKR more efficiently, or that it is more resistant to RNase L, or has a generally increased rate of translation initiation and/or elongation. It also remains to be shown whether m1 Ψ containing stop codons have an increased rate of translational read through as previously shown for Ψ containing stop codons [33,34].

In conclusion, in this manuscript, we show that (m5C)/m1 Ψ modified mRNA was superior to (m5C)/ Ψ containing mRNA in its capacity to produce protein and its ability to reduce the intracellular innate immune response. In a complementary study, Pardi et al. [35]

(accepted while this manuscript was under review) characterized how various RNA delivery routes affected in vivo protein expression kinetics using m1 Ψ modified mRNA. Future work may demonstrate the benefits of using m1 Ψ modified mRNA for medical applications such as genetic reprogramming, stem cell differentiation, protein replacement therapy, vaccination, passive immunoprophylaxis (vectored antibody expression), and other advanced therapeutic goals [36].

Acknowledgments

O. Andries was supported by a PhD fellowship and international mobility grant from Fonds Wetenschappelijk Onderzoek (FWO) and the Emmanuel van der Schueren fellowship from Vlaamse Liga tegen Kanker (VLK). This work was supported by grants from the Defense Advanced Research Projects Agency (DARPA) and an innovation award from the Ragon Institute of MGH, MIT and Harvard to R. Weiss, Bijzonder Onderzoeksfonds (BOF) from Ghent University and grants from FWO (G.0235.11 N and G.0621.10 N) to N. N. Sanders.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.jconrel.2015.08.051>.

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